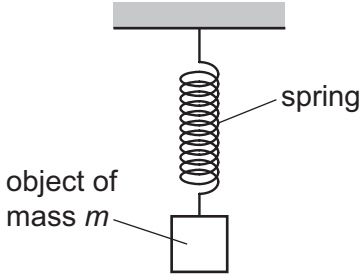
3 An object of mass m is suspended by a spring from a fixed point.



The spring has spring constant k. The object is set into vertical oscillations of period T.

Which equation for T is homogeneous with respect to base units?

A $T = 2\pi\left(\frac{k}{m}\right)$ **B** $T = 2\pi\left(\frac{m}{k}\right)$ **C** $T = 2\pi\sqrt{\frac{k}{m}}$ **D** $T = 2\pi\sqrt{\frac{m}{k}}$